TOWARDS A DICHOTOMY FOR THE LIST SWITCH HOMOMORPHISM PROBLEM FOR SIGNED GRAPHS

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Abstract. We make advances towards a structural characterisation of the signed graphs H for which the list switch H-colouring problem LSwHom(H) problem is polynomial time solvable. We conjecture a characterisation for signed graphs that can be switched to graphs such that every negative edge is also positive, and prove the characterisation in the case that the signed graph is reflexive.

1. Introduction

The CSP-dichotomy of Bulatov [5] and Zhuk [16] tells us that the constraint satisfaction problem CSP($\mathcal{H}$) for a core relational structure $\mathcal{H}$ is polynomial time solvable (in P) if $\mathcal{H}$ admits a WNU-polymorphism, and is otherwise NP-compete (in NPC). It is difficult, however, to decide if $\mathcal{H}$ admits a WNU-polymorphism. So more tractable dichotomies, characterisations like the well known H-colouring dichotomy of [11] which says that the homomorphism problem Hom(H) for a simple graph H is in P if and only if H is bipartite, are still sought. One such dichotomy arose recently in relation to signed graphs, and in this paper we work towards a generalisation of this. The definition of WNU-polymorphism, as well as the definitions of many other now standard algebraic terms introduced in this section without definition, can be found in Section 2.

Henceforth, *graphs* are undirected graphs in which loops are allowed. If all vertices have loops then the graph is *reflexive*, and if no vertices have loops then it is *irreflexive*. We denote an edge $\{u, v\}$ of a graph simply as uv, and write $u \sim v$ to mean that uv is an edge.

A *signed graph* is a graph G together with an assignment of a sign + or − to each edge. Introduced by Harary in [10] in 1955, there are numerous results about signed graphs. For our purposes, it will be enough to view a signed graph G simply as a br-*graph* (for blue-red): a pair of graphs– G^B, whose edges are called *blue edges*, and G^R, whose edges are *red edges*– on the same vertex set. Edges with both colours are called *purple edges*. If all edges are red (or purple) we call the br-graph *red* (or *purple*). If an edge is red (or blue) but not purple, we call it *pure* red (or pure blue). A br-graph G is irreflexive, or bipartite, or connected or such, if the *underlying graph* $G^R \cup G^B$ is. Many results about signed graphs deal with the operation of *switching*, introduced by Zaslavsky [15], in which, for a vertex v,

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one switches between G^R and G^B the sets of edges incident to v, and considers the class of br-graphs one can get from G via switches. Such graphs are called *switch-equivalent* to G. A br-graph that is switch equivalent to a graph with no pure blue edges is a pr-*graph* (for purple-red).

As a *homomorphism* $\phi : G \to H$ of a graph G to a graph H is a edge preserving vertex map, a *homomorphism* from a br-graph G to another br-graph H is a vertex map $\phi : G \to H$ which preserves both red edges and blue edges. Defined in [14], a *switch-homomorphism* from G to H is a vertex map $\phi : G \to H$, such that for some graph G' that is switch-equivalent to G, $\phi : G' \to H$ is a homomorphism.

In [9], Foucaud and Naserasr introduced the decision problem $\mathsf{SwHom}(H)$ for a given br-graph H: decide for a given br-graph G if there is a switch-homomorphism to H. They addressed the problem of classifying the complexity of $\mathsf{SwHom}(H)$ in terms of H. A priori, $\mathsf{SwHom}(H)$ is not a CSP problem, but in [3] the authors showed that it is polynomially equivalent to $\mathsf{CSP}(\mathcal{S}_H)$ for a br-graph $\mathcal{S}_H$ called the *switching graph* of H, defined in Definition 2.3. From this, it follows by the CSP-dichotomy, that there is a complexity dichotomy. This dichotomy was characterised in [3] and [2].

The *switch-core* of a br-graph H is the unique minimal induced subgraph to which it admits a switch-homomorphism. The following theorem, conjectured in [3], was completed in [2].

Theorem 1.1 ([2])**.** *For a* br-*graph H, the problem* $\mathsf{SwHom}(H)$ *is in* P *if the switch-core of H has at most two edges, and is otherwise in* NPC.

Our goal is to determine, for a given br-graph H, the complexity of the list version $\mathsf{LSwHom}(H)$ of the problem $\mathsf{SwHom}(H)$.

Much has already been done in [1]. Building on known ideas that we explain in more detail in the next section, the authors first observe that $\mathsf{LSwHom}(H)$ is in P if and only if $\mathcal{S}_H$ admits a *semi-conservative* WNU-polymorphism, appropriately defined at the end of Section 2. The goal then becomes to characterise the br-graphs H such that $\mathcal{S}_H$ admits a semi-conservative WNU-polymorphism. The authors then go on to define a special bipartite-min ordering of a graph H and two obstructions to its existence– invertible pairs and chains, both of which are detectable in polynomial time. Using these as their main tools, they characterise the br-trees admitting semi-conservative WNU-polymorphisms. The full characterisation is not easy to state, but in the restricted case that H is a irreflexive trees, it has a simpler statement.

It is a simple task to show that an irreflexive br-tree is a pr-graph: working from a root outwards one can switch all pure blue edges to pure red. In [1], the authors prove the following in the case that the pr-graph H is a tree. (The definitions of invertible pairs, chains, and special bipartite-min orderings of pr-graphs are given at the beginning of Section 5.)

Conjecture 1.2. *For a* pr-*graph H, the following are equivalent.*

(i) H has no invertible pairs no chains.
(ii) H has a special bipartite-min ordering.
(iii) $\mathcal{S}_H$ has a semi-conservative WNU*-polymorphism.*

Our main results is to prove this in the case that H is a reflexive pr-graph. Though irreflexive trees are pr-graphs, reflexive trees are not. So this does not strictly generalise even the reflexive results of [1]. That said, br-trees are 'almost' pr-graphs in that the only pure blue edges that one cannot 'switch away' are blue

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loops. These blue loops add considerable complication to the proofs of in [1], and it is evident from examples in [1] that Conjecture 1.2 will not generalise to br-graphs without some alternation.

Though we do not generalise the results of [1], our result seems to be a significant complement to theirs, not only in the classes of graphs dealt with, but also in the techniques used. Our proofs make significant use of symmetries both in the definition of $\mathcal{S}_H$, and in the definition of bipartite-min orderings.

Section 2 can be viewed as, perhaps, an expansion of this introductory section. We recall many required algebraic results of CSP theory and structural results about the list homomorphism problem for graphs.

In Section 3 we define a switch-symmetric polymorphism which commutes with the obvious symmetry in $\mathcal{S}_H$, and use it to observe that if a polymorphism of the red subgraph $\mathcal{S}_H^R$ of $\mathcal{S}_H$ is switch-symmetric, then it is a polymorphism of $\mathcal{S}_H$. This is a useful tool in defining polymorphisms on $\mathcal{S}_H$, but also yield a quick result. It allows to quickly prove Theorem 3.2 which characterises the br-graphs H such that $\mathcal{S}_H$ admits a conservative WNU-polymorphism. This depends on known characterisations of graphs with conservative WNU-polymorphisms. As these do not exist for semi-conservative WNU-polymorphisms though, much work remains.

In Section 4 we make observations about a 'reduction to the bipartite case' for list homomorphisms of graphs that is given in [8], and show that parts of it extend transparently to br-graphs. In particular, this leads to the definition of a 'parity-symmetric' bipartite-min ordering, and the following strengthening of Conjecture 1.2.

Conjecture 1.3. *For a* pr*-graph H, the following are equivalent.*

(i) H has no invertible pairs or chains.
(ii) H has a parity-symmetric special bipartite-min ordering.*
(iii) $\mathcal{S}_H$ has a semi-conservative WNU*-polymorphism.*

As $(ii*) \Rightarrow (ii)$ is trivial and $(ii) \Rightarrow (i)$ is shown in [1] this is indeed a stronger version of Conjecture 1.2. In fact it consists of Conjecture 1.2 and the following, the proof of which, for graphs, follows from non-trivial results in [8].

Conjecture 1.4. *For a* pr*-graph H, $\mathcal{S}_H$ has a special bipartite-min ordering if and only if it has a parity-symmetric one.*

In Sections 5 and 6 we prove, as Theorem 5.3, Conjecture 1.3 for reflexive pr-graphs. This also proves Conjecture 1.4 for reflexive pr-graphs. In Section 5 we observe that because we are restricted to pr-graphs, $(iii) \Rightarrow (i)$ follows immediately from results of [1]. We then prove $(ii*) \Rightarrow (iii)$ for all pr-graphs, (proving first $(ii) \Rightarrow (iii)$) of Conjecture 1.2 for bipartite pr-graphs). The main step, which comes in Section 6, is proving $(i) \Rightarrow (ii*)$ for reflexive pr-graphs.

The literature about the list homomorphism problem that we recall in Section 2 suggests many 'polymorphism collapses' that may be useful finding a full characterisation of the complexity of $\mathsf{LSwHom}(H)$. While some have proved useful, some have been red herrings. In our final section, Section 7 we give examples showing that some of these collapses do not occur.

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2. Background

In this section we recall the definitions required to properly state our problem, and make some initial observations about it. Many of these observations can be found in [1].

2.1. Structures, CSP, and List Homomorphism.

A (relational) structure $\mathcal{H}$ consists of a set $V = V(\mathcal{H})$ of vertices with a finite ordered set of finite arity relations $R_i \subset V^k$ on V. The corresponding ordered set of the arities of the structure is called its *type*. A *homomorphism* between two structures of the same type is a vertex map that preserves each relation. The problem $\mathsf{CSP}(\mathcal{H})$ for a structure $\mathcal{H}$ is the problem of deciding if an instance structure $\mathcal{G}$ of the same type admits a homomorphism to $\mathcal{H}$.

A graph H is viewed as a structure with one symmetric binary relation E, an edge uv corresponding to the pairs (u, v) and (v, u) in E. A vertex map $V(G) \to V(H)$ of graphs is a homomorphism between two graphs if and only if it is a homomorphism of the corresponding symmetric binary structures. So $\mathsf{Hom}(H)$ is polynomially equivalent to $\mathsf{CSP}(H)$.

In the list variation $\mathsf{LHom}(H)$ of the homomorphism problem $\mathsf{Hom}(H)$, an instance G comes attached with a list $L(v) \subset V(H)$ for each vertex $v \in V(G)$, and one must decide if there is a homomorphism $\phi : G \to H$ such that $\phi(v) \in L(v)$ for each vertex v of G. This is equivalent to $\mathsf{CSP}(H_c)$ where $\mathcal{H}_c$ is the relational structure one gets from a structure $\mathcal{H}$ by adding a new unary relation $L_S = \{(s) \mid s \in S\}$ for each subset $S \subset V(\mathcal{H})$.

The *core* $\mathcal{C}(\mathcal{H})$ of a structure $\mathcal{H}$ is the unique minimum induced substructure to which it admits a homomorphism. It is basic that $\mathsf{CSP}(\mathcal{C}(\mathcal{H}))$ is polynomially equivalent to $\mathsf{CSP}(\mathcal{H})$.

A k-ary polymorphism ϕ of $\mathcal{H}$ is a homomorphism $\phi : \mathcal{H}^k \to \mathcal{H}$ (that is, a map preserving all relations of $\mathcal{H}$), where $\mathcal{H}^k$ is the k-time categorical product of $\mathcal{H}$ with itself. It is *idempotent* if for all $h \in V(\mathcal{H})$, $\phi(h, h, \ldots, h) = h$ and it is *conservative* if $\phi(v_1, \ldots, v_k) \in \{v_1, \ldots, v_k\}$. An idempotent polymorphism $\phi : \mathcal{H}^k \to \mathcal{H}$, for $k \geq 3$ is *weak near unanimity* or WNU if for every choice of $x, y \in V(\mathcal{H})$ we have

$$\phi(x, x, \ldots, x, y) = \phi(x, x, \ldots, y, x) = \ldots \phi(y, x, \ldots, x, x).$$

The following is proved in [5] and [16].

Theorem 2.1 (The CSP-dichotomy)**.** *For a relational structure $\mathcal{H}$ the problem $\mathsf{CSP}(\mathcal{H})$ is in P if the core of $\mathcal{H}$ admits a WNU-polymorphism, and is otherwise in NPC.*

In general it is difficult to find the core of a structure $\mathcal{H}$, and tends to be difficult to give a non-algebraic description of the structures $\mathcal{H}$ for which $\mathsf{CSP}(\mathcal{H})$ is in P. For list colouring, it gets a little easier. For one, the structure H_c is always a core. It is easy to see that any polymorphism of H_c is conservative, so for list colouring problems we get the following corollary of Theorem 2.1 which Bulatov actually proved years earlier in [4].

Corollary 2.2. *For a relational structure $\mathcal{H}$ the problem $\mathsf{CSP}(\mathcal{H}_c)$ is in P if and only if the core of $\mathcal{H}$ admits a conservative WNU-polymorphism.*

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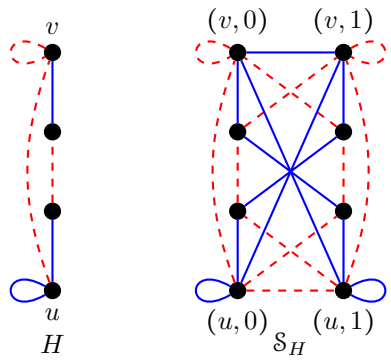

FIGURE 1. A br-graph H and its switching graph $\mathcal{S}_H$.

2.2. **The switching graph.** One might observe that $\mathsf{SwHom}(H)$ for a br-graph H is not a CSP, so the above tools do not apply, but one of the main tools in [3] and [2] is the following construction which allows us to view $\mathsf{SwHom}(H)$ as a CSP.

For a product $A \times B$ of sets or of graph, we use π_A for the projection map onto A and π_B for the projection map onto B.

Definition 2.3. Let H be a br-graph, and S be the set $\{0,1\}$. The *switching graph* $\mathcal{S}_H$ of H is the br-graph on the set $V(H) \times S$, such that each of the sets $\pi_S^{-1}(i) = \{(v,i) \mid v \in v(H)\}$ induces a copy of H (identified through the projection π_H in the obvious way), and such that, where the *switch* map $f_s : V(\mathcal{S}_H) \to V(\mathcal{S}_H)$ is defined $f_s((v,i)) = (v, 1-i)$, the edge $uf_s(v)$ is red if uv is blue, and vice-versa.

See Figure 1 for an example of the switching graph construction. (In all figures, dashed edges are red, dotted edges are non-edges.) Blue edges are solid, while red edges are dashed. The following tells us that $\mathsf{SwHom}(H)$ is polynomially equivalent to the homomorphism problem $\mathsf{Hom}(\mathcal{S}_H)$ for the br-graph $\mathcal{S}_H$.

Proposition 2.4 ([3])**.** *For* br*-graphs G and H, G admits a switch-homomorphism to H if and only if it admits a homomorphism to $\mathcal{S}_H$.*

2.3. **Definition of List Switch-Homomorphism.** Though the problems $\mathsf{SwHom}(H)$ and $\mathsf{Hom}(\mathcal{S}_H)$ are polynomially equivalent, they have different list versions. For a br-graph H we define the following two problems.

$\mathsf{LSwHom}(H)$
Instance: A br-graph G with lists $L : V(G) \to 2^{V(H)}$.
Question: Does G admit a switch-homomorphism $\phi : G \to H$ preserving lists?

$\mathsf{LHom}(\mathcal{S}_H)$
Instance: A br-graph G with lists $L : V(G) \to 2^{V(\mathcal{S}_H)}$.
Question: Does G admit a homomorphism $\phi : G \to \mathcal{S}_H$ preserving lists?

The second problem is clearly equivalent to the CSP problem $\mathsf{CSP}(\mathcal{S}_H^c)$. The first problem can also be made equivalent to a CSP using Proposition 2.4. Indeed, from the proof of this proposition, it is not hard to that it is polynomially equivalent to $\mathsf{CSP}(\mathcal{S}_H^{sc})$ where we get $\mathcal{S}_H^{sc}$ from $\mathcal{S}_H^c$ by throwing away non-symmetric lists, i.e.,

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throwing away the unary relation L_S for any subset $S \subset V(H)$ that is not closed under action by the switch map s from Definition 2.3.

Clearly $\mathsf{LHom}(\mathcal{S}_H)$ encodes $\mathsf{LSwHom}(H)$ by taking only instances with symmetric lists, and as the full lists are symmetric, $\mathsf{LSwHom}(H)$ encodes $\mathsf{Hom}(\mathcal{S}_H)$ by taking instances with full lists on every vertex. Thus we have the following polynomial reductions.

$$\mathsf{Hom}(\mathcal{S}_H) \leq_{\text{poly}} \mathsf{LSwHom}(H) \leq_{\text{poly}} \mathsf{LHom}(\mathcal{S}_H). \tag{1}$$

The relationship between them has a nice algebraic specification. Calling a polymorphism of $\mathcal{S}_H$ semi-conservative if it conserves symmetric lists, Theorem 2.1 yields the following.

Fact 2.5. *For a* br*-graph H,* $\mathsf{LSwHom}(H)$ *is in* P *if and only if $\mathcal{S}_H$ admits a semi-conservative* WNU*-polymorphism, and* $\mathsf{LHom}(\mathcal{S}_H)$ *is in* P *if and only if $\mathcal{S}_H$ admits a conservative* WNU*-polymorphism.*

Proof. Everything would be immediate from Theorem 2.1 except that it applies to cores, and $\mathcal{S}_H^{sc}$ (which we use for $\mathsf{LSwHom}(H)$) may not be a core. So we have to observe that if $\mathcal{S}_H$ admits a semi-conservative WNU-polymorphism then the core of $\mathcal{S}_H^{sc}$ admits a WNU-polymorphism. To get the core of $\mathcal{S}_H^{sc}$ we must identify the vertices $(v, 0)$ and $(v, 1)$ for any vertex v of H all of whose edges are purple. Let r be the retraction of $\mathcal{S}_H$ to its core that makes all of these identifications. It is easy to check that if $\phi : \mathcal{S}_H^k \to \mathcal{S}_H$ is semi-conservative WNU then $\phi' = r \circ \phi|_{r(\mathcal{S}_H)^k}$ is a WNU-polymorphism on the core $r(\mathcal{S}_H)$. □

2.4. **Collapses of Conservative Polymorphisms.** Our goal is a nice characterisation of the br-graphs H such that $\mathcal{S}_H$ admits a (semi-)conservative WNU-polymorphism. In the case of other graph-like structures, such characterisations often come from the fact that such structures have conservative WNU-polymorphisms only if they have those that satisfy certain stronger identities that are known to be related to nice structural descriptions.

The first such conservative WNU-polymorphisms that we mention is the 3-NU-polymorphism, or the majority polymorphism.

Definition 2.6. A polymorphism $\phi : H^3 \to H$ on a graph H is a 3-NU if for any $x, y \in V(H)$, we have

$$\phi(x, x, y) = \phi(x, y, x) = \phi(y, x, x) = x.$$

It is clear that a 3-NU-polymorphism is WNU, but one can find many examples of structures with WNU-polymorphisms that do not have even k-NU-polymorphisms. However, the following collapse of these two types of polymorphism comes from [8].

Theorem 2.7. *A graph H has a conservative* WNU*-polymorphism if and only if and only if it has a conservative* 3-NU*-polymorphism.*

The other Conservative WNU-polymorphism we use is disguised as an ordering.

Definition 2.8. An ordering $\leq$ of the vertices of a reflexive graph is a *min ordering* if the 2-ary min operation $\min(u, v)$, defined by $\min(u, v) = u$ if $u \leq v$, is a polymorphism.

A graph with min ordering admits the induced 3-ary operation $\min(a, b, c) = \min(\min(a, b), c)$ which is a conservative WNU-polymorphism of H; we call this the

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min-polymorphism associated with the min ordering. From [6], where the $\mathsf{LHom}(H)$ dichotomy was given for reflexive graphs H, we get the following.

Theorem 2.9. *For a reflexive graph H the following are equivalent.*

- *H admits a conservative* WNU*-polymorphism.*
- *H admits a min ordering.*
- *H contains no invertible pairs.*

This is extremely useful because the existence of min orderings is also known to characterise the well known class of interval graphs, so a reflexive graph has a conservative WNU-polymorphism if and only if it is an interval graph. Min orderings are also known to be characterised by the fact that they satisfy the following, so-called, *underbar* identity:

$$(a \le a', b \le b', a \sim b', a' \sim b) \Rightarrow (a \sim b). \tag{2}$$

No non-empty irreflexive graph can have a min ordering, but fortunately there is a variation on this useful characterisation that works for all graphs. A bipartite-min ordering, defined in Section 4 is a generalisation of a min-ordering for reflexive graphs, and exists on many irreflexive graphs.

It was shown in [7] that a for a graph H with a bipartite-min ordering, the problem $\mathsf{LHom}(H)$ can be solved in polynomial time via the arc-consistency algorithm. In [13] the authors generalised the notion of bipartite-min orderings to digraphs, unifying many known list homomorphism characterisations. The following generalisation of Theorem 2.9 is from [13].

Theorem 2.10. *[13] For a graph H the following are equivalent.*

- *H admits a conservative* WNU*-polymorphism.*
- *H admits a bipartite-min ordering.*
- *H contains no invertible pairs.*

3. Dichotomy for List $\mathcal{S}_H$-colouring

Recall the switch automorphism $f_s : \mathcal{S}_H \to \mathcal{S}_H$ that maps (v, i) to $(v, 1-i)$. This is clearly also an automorphism of $\mathcal{S}_H^R$. A k-ary polymorphism ϕ of $\mathcal{S}_H$, or of $\mathcal{S}_H^R$, is *switch-symmetric* if it commutes with f_s:

$$\phi(f_s(v_1), \dots, f_s(v_k)) = f_s(\phi(v_1, \dots, v_k)).$$

The following is easy.

Lemma 3.1. *If a polymorphism of $\mathcal{S}_H^R$ is switch-symmetric, then it is a polymorphism of $\mathcal{S}_H$.*

Proof. Let $\phi : (\mathcal{S}_H^R)^k \to \mathcal{S}_H^R$ be switch-symmetric. To see that it is a polymorphism of $\mathcal{S}_H$ it is enough to observe that it is also a polymorphism of $\mathcal{S}_H^B$. Let $x_i \sim y_i$ in $\mathcal{S}_H^B$ for each $i \in [k]$. By the definition of $\mathcal{S}_H$ we have that $x_i \sim f_s(y_i)$ in $\mathcal{S}_H^R$ for each i. So as ϕ is a polymorphism of $\mathcal{S}_H^R$ we get

$$\phi(x_1, \dots, x_k) \sim \phi(f_s(y_1), \dots, f_s(y_k)) = f_s(\phi(y_1, \dots, y_k))$$

is a red edge, and so $\phi(x_1, \dots, x_k) \sim \phi(y_1, \dots, y_k)$ is a blue edge. This is enough. □

We will use this in Section 5, but for now we exhibit its utility by using it to characterise the graphs $\mathcal{S}_H$ having conservative WNU-polymorphisms. This yields a simple complexity dichotomy for the problem $\mathsf{LHom}(\mathcal{S}_H)$.

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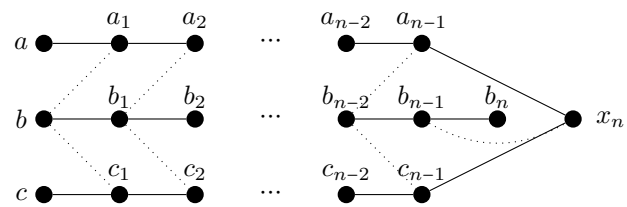

FIGURE 2. A b-excluder for (a, b, c)

We show the following.

Theorem 3.2. *For a* br*-graph H, $\mathcal{S}_H$ admits a conservative* WNU*-polymorphism if and only if $\mathcal{S}_H^R$ admits a conservative* 3-NU*-polymorphism.*

The proof is mostly observations about known results. First observe the following hierarchy of polymorphisms.

$$\begin{aligned} \mathcal{S}_H \text{ has a conservative 3-NU} &\Rightarrow \mathcal{S}_H \text{ has a conservative WNU} \\ &\Rightarrow \mathcal{S}_H^R \text{ has a conservative WNU} \\ &\Rightarrow \mathcal{S}_H^R \text{ has a conservative 3-NU} \end{aligned} \tag{3}$$

The first two implications are trivial; indeed, the first is by definition, and the second is because a polymorphism of a structure is a polymorphism of any relation of the structure. The third, we get by applying Theorem 2.7 to the graph $\mathcal{S}_H^R$.

To prove Theorem 3.2 it is enough to show that a 3-NU-polymorphism on $\mathcal{S}_H^R$ may be assumed to by switch-symmetric, as then be Lemma 3.1 we get that the four statements in (3) are equivalent. In fact one can assume more about the 3-NU-polymorphism on $\mathcal{S}_H^R$

Definition 3.3. A conservative 3-NU-polymorphism $\phi : H^3 \to H$ is *symmetric* if it commutes with all automorphisms σ of H:

$$\phi(\sigma(a), \sigma(b), \sigma(c)) = \sigma(\phi(a, b, c)).$$

In [12], while giving a characterisation of the digraphs H with tractable $\mathsf{LHom}(H)$, Hell and Rafiey found a useful omitted-subgraph characterisation of digraphs with conservative NU-polymorphisms. Their definition is for digraphs, but we give it here for graphs.

Definition 3.4. A walk $P = u_0 \sim u_1 \sim \cdots \sim u_n$ of a graph H *avoids* a walk $Q = v_0 \sim v_1 \sim \cdots \sim v_n$ of H of the same length if $u_{i-1} \not\sim v_i$ for each $i \in [n]$. For a 3-tuple (a, b, c) of vertices of H, a b-*excluder* in H is a set of three walks B_a, B_b and B_c of the same length, starting at a, b and c respectively, such that B_a and B_c share the same last vertex, and B_b avoids B_a and B_c. See Figure 2. A 3-tuple (a, b, c) is a *permutable triple* if H contains an a-excluder, a b-excluder, and a c-excluder.

They showed the following.

Theorem 3.5. *[12] A graph H has a* 3-NU*-polymorphism if and only if it contains no permutable triples.*

If a graph H has no permutable triples then for any 3-tuple (x_1, x_2, x_3) in $V(H)^3$, there is at least one $i \in [3]$ such that there is no x_i-excluder in H for (x_1, x_2, x_3). The following is in the proof of Theorem 4.1 of [12].

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Lemma 3.6. *Let H be a graph containing no permutable triples. The function $\phi : H^3 \to H$ defined as follows is a conservative* 3-NU*-polymorphism of H. Let $\phi(x_1, x_2, x_3) = m$ if at least two of the entries are m, and otherwise let $\phi(x_1, x_2, x_3) = x_i$ where i is the minimum index such that there is no x_i-excluder in H for (x_1, x_2, x_3).*

It is clear that existence of excluders for a triple is preserved under any automorphism s of H. The properties 'two of x_1, x_2 and x_3 being the same', and 'minimum index' are also preserved, so the conservative 3-NU-polymorphism defined in the lemma is symmetric.

This allow us to prove Theorem 3.2.

Proof of Theorem 3.2. Let $\mathcal{S}_H^R$ have a conservative 3-NU-polymorphism ϕ. By the discussion following Lemma 3.6 we may assume that ϕ is symmetric, and so, as the switch map is an automorphism of $\mathcal{S}_H^R$, switch-symmetric. By Lemma 3.1, ϕ is then conservative 3-NU-polymorphism of $\mathcal{S}_H$, and so a conservative WNU-polymorphism. This completes the prove of the equivalence of the four polymorphisms in (3), and so proves Theorem 3.2. □

4. Bipartite-min orderings and reduction to bipartite graphs

In [8], Feder, Hell, and Huang, reduced the complexity of the list homomorphism problem for graphs to the bipartite (so irreflexive) case by showing that $\mathsf{LHom}(H)$ for any graph H is polynomially equivalent to $\mathsf{LHom}(\mathcal{B}_H)$, for a bipartite graph $\mathcal{B}_H := K_2 \times H$. Their result implies that H has a conservative WNU-polymorphism if and only if $\mathcal{B}_H$ does. Their proof consists of two parts. Where 'parity-symmetric' polymorphism on $\mathcal{B}_H$ are defined below, then essentially prove the following two equivalences.

(i) H has conservative WNU-polymorphism if and only if $\mathcal{B}_H$ has a parity-symmetric conservative WNU-polymorphism.
(ii) $\mathcal{B}_H$ has a parity-symmetric conservative WNU-polymorphism if and only if it has a parity-symmetric WNU-polymorphism.

(To reconcile these statements with those of [8], the reader should observe that the graph H^{**} of [8] is simply $\mathcal{B}_H$ augmented with a relation that forces all polymorphisms to be parity-symmetric.)

The first statement is easy. In this section we observe that it generalises and extends to similar statements about polymorphisms of br-graphs, and in particular about polymorphisms of $\mathcal{S}_H$. The second statement is much harder and depends on structural results.

4.1. The bipartite resolution and the constant parity subgraph.

Recall that for a graph H, the categorical product $K_2 \times H$ is the graph with vertex set $\{0,1\} \times V(H)$ in which $(i,h) \sim (i',h')$ if and only if $i \neq i'$ and $h \sim h'$.

Let B be the purple K_2 on the vertex set $\{0,1\}$. For a br-graph H, the *bipartite resolution* of H is $\mathcal{B}_H := B \times H$. It is a bipartite br-graph with partite sets $\pi_B^{-1}(0)$ and $\pi_B^{-1}(1)$, so we call $i = \pi_B((i,v))$ the *parity* of a vertex (i,v) and call the map

$$f_p : (i,v) \mapsto (1-i, v), \tag{4}$$

which is clearly an automorphism of $\mathcal{B}_H$, the *parity flip* map. A polymorphism ϕ of $\mathcal{B}_H$ is *parity-symmetric* if it commutes with the parity flip map.

Let $\mathcal{P}$ be the set consisting of the following properties applying to polymorphisms of a graph H:

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idempotent, conservative, semi-conservative (if H is a switch graph), k-NU, WNU.

For any subset $P \subseteq \mathcal{P}$, a polymorphism ϕ of H is a P-polymorphism if it has all properties in P. The main theorem of this section is the following.

Lemma 4.1. *For any* br*-graph H, and any subset P of $\mathcal{P}$, H has a P-polymorphism if and only if $\mathcal{B}_H$ has a parity-symmetric P-polymorphism.*

Proof. This is immediate from Claims 4.2 and 4.3 given below. □

For a vertex $v = (v_1, \dots, v_k)$ in $\mathcal{B}_H^k$, the *parity pattern* is the tuple

$$\pi_B(v) := (\pi_B(v_1), \dots \pi_B(v_k))$$

of its coordinates' parities. Over components of $\mathcal{B}_H$ the parity pattern is constant up to component-wise action of the parity flip map f_p. In particular the *constant parity subgraph* $\mathrm{CP}(\mathcal{B}_H^k)$ of $\mathcal{B}_H^k$, which is the subgraph induced on the vertices with parity patterns $(0, 0, \dots, 0)$ or $(1, 1, \dots, 1)$, is a union of components. A coordinate of a parity pattern is *majority* if at least half of the coordinates have the same parity. A homomorphism $\Phi : \mathrm{CP}(\mathcal{B}_H^k) \to \mathcal{B}_H$ is a P-homomorphism, for any $P \subseteq \mathcal{P}$ if it satisfies those same conditions required of a polymorphism of $\mathcal{B}_H$ to satisfy them, and is parity-symmetric if it commutes with f_p.

Claim 4.2. *For any* br*-graph H and any $P \subseteq \mathcal{P}$, H has a k-ary P-polymorphism if and only if $\mathrm{CP}(\mathcal{B}_H^k)$ has a parity-symmetric P-polymorphism.*

Proof. The details of the proof are straight-forward but tedious, and so we just outline the main points.

For a polymorphism $\phi : H^k \to H$, the map $\Phi : \mathrm{CP}(\mathcal{B}_H^k) \to \mathcal{B}_H$ defined for $v = (v_1, \dots, v_k) \in V(\mathrm{CP}(\mathcal{B}_H^k))$ by

$$\Phi(v) = (\pi_B(v_1), \phi(\pi_H(v_1), \dots \pi_H(v_k)))$$

is a parity-symmetric homomorphism, and for a parity-symmetric homomorphism $\Phi : \mathrm{CP}(\mathcal{B}_H^k) \to \mathcal{B}_H$ the map $\phi : H^k \to H$ defined for $v = (v_1, \dots, v_k)$ by

$$\phi(v) = (\Phi((0, v_1), \dots, (0, v_k))$$

is a polymorphism; moreover, these constructions are inverse.

That ϕ is WNU or k-NU if and only if Φ is is also straight-forward. To verify that ϕ is idempotent, conservative, or semi-conservative if and only if Φ is one need only observe that Φ preserves a list $L \subset V(\mathcal{B}_H)$ if and only if ϕ preserves its intersections with each side of $\mathcal{B}_H$: $\pi_H(L \cap \pi_B^{-1}(0))$ and $\pi_H(L \cap \pi_B^{-1}(1))$. □

Claim 4.3. *For any* br*-graph H and any $P \subseteq \mathcal{P} \cup \{$parity-symmetric$\}$, any P-homomorphism $\mathrm{CP}(\mathcal{B}_H^k) \to \mathcal{B}_H$ extends to a P-polymorphism of $\mathcal{B}_H$.*

Proof. Given a homomorphism $\Phi : \mathrm{CP}(\mathcal{B}_H^k) \to \mathcal{B}_H$ extend it to a polymorphism of $\mathcal{B}_H$ by defining it, on each component of $\mathrm{CP}(\mathcal{B}_H^k)$, to project onto some coordinate that is majority in the parity pattern of the component.

That this maintains the properties of idempotence, conservativity, and semi-conservativity is immediate from the observation that projections preserve all lists. That parity-symmetry is maintained is immediate from the fact that projections are parity-symmetric. That the properties k-NU and WNU are maintained uses the choice of projection, but it straight-forward. □

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We finish this subsection with one more simple observation.

Fact 4.4. *For any* br*-graph* H*,* $\mathcal{B}_{\mathcal{S}_H}$ *and* $\mathcal{S}_{\mathcal{B}_H}$ *are isomorphic.*

Proof. They are both the graph with vertex set $B \times V(H) \times S$ where $(i,h,j) \sim (i',h',j')$ is an edge if and only if $i \neq i'$ and $h \sim h'$. The edge is red if and only if $h \sim h'$ is red and $j = j'$ or $h \sim h'$ is blue and $j \neq j'$. □

4.2. **Bipartite-min orderings.** Bipartite-min orderings were first defined on bipartite graphs as an analogue of min orderings.

Definition 4.5. An ordering of the vertices of a bipartite graph H with partite sets U and V is a *bipartite-min ordering* if for any $a, b \in U$ and $a', b' \in V$ the, underbar identity (2) holds.

The extension of the definition of bipartite-min orderings to general graphs is simple.

Definition 4.6. A *bipartite-min ordering of* H is a bipartite-min ordering of $\mathcal{B}_H$

Strictly speaking, this definition should only be made for non-bipartite graphs, however if H is bipartite then $\mathcal{B}_H$ is just two disjoint copies of H, and we can ignore one of those copies with impunity. This has the added notational expedience of allowing us to assume that bipartite graphs are of the form $\mathcal{B}_H$.

Though a bipartite-min ordering of $\mathcal{B}_H$ is an ordering of $V(\mathcal{B}_H)$, it is often useful to take the equivalent view of it as two orderings, one on each of the bipartite sets $\pi_B^{-1}(0)$ and $\pi_B^{-1}(1)$. We say a such a bipartite-min ordering is *parity-symmetric* if the orderings on these sets are the same.

While we observed in the introduction that a min orderings of H defines a conservative WNU-polymorphism $\min : H^3 \to H$, the same it not true of bipartite-min orderings on $\mathcal{B}_H$. However, it is easy to check that the min function it defines is a conservative WNU-homomorphism from $\mathrm{CP}(\mathcal{B}_H^3)$ to $\mathcal{B}_H$, so by Claim 4.3 extends to a conservative WNU-polymorphism of $\mathcal{B}_H$. We refer to this as a bipartite-min polymorphism of $\mathcal{B}_H$. If the bipartite-min ordering on $\mathcal{B}_H$ is parity-symmetric, then so is this bipartite-min polymorphism of $\mathcal{B}_H$, so by Lemma 4.1 gives a conservative WNU-polymorphism on H.

It follows that a parity-symmetric bipartite-min ordering on H puts $\mathsf{LHom}(H)$ in P. In [8], they accomplished the harder task of showing the converse statement.

5. Special bipartite-min orderings and purple-red graphs

For the remainder of the paper, we consider pr-graphs, and we will assume that they have been switched to remove all pure blue edges. We thus talk of red edges, by which we mean pure red edges, and purple edges. In the case that we want to consider a purple edge as red, say in the target of a homomorphism, we will say 'red or purple' or 'not necessarily pure red'.

The following definitions have more general versions in [1] but we give simplified versions for pr-graphs.

Definition 5.1. Given a pr-graph H with a bipartite-min ordering of the underlying graph, a vertex is *special* if all its neighbours via purple edges come before (below) its other neighbours. The ordering is a *special bipartite-min ordering* if all vertices are special.

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Definition 5.2. Let H be a pr-graph. Construct a digraph $\mathcal{A}$ on the vertex set $V(H)^2$ by letting $(a,b) \to (a',b')$ if $a \sim a'$, $b \sim b'$ and $a \not\sim b'$. A pair (a,b) is a *invertible pair* if there is a walk from (a,b) to (b,a) in $\mathcal{A}$. A *good-end* is a pair (a,b) of vertices of H for which there is another vertex x that with a red edge to b and a purple edge to a. A *bad-end* is a pair (a,b) for which there is another vertex y with a red edge to a and a purple edge to b. A *chain* in H is a directed walk in $\mathcal{A}$ from a good-end to a bad end.

Note that this definition of an invertible pair in H corresponds with the known definition of an invertible pair in the underlying graph of H.

Our main theorem is Conjecture 1.3 for reflexive pr-graphs.

Theorem 5.3. *For a reflexive* pr*-graph H, the following are equivalent.*

(i) H has no invertible pairs or chains.
(ii) H has a parity-symmetric special bipartite-min ordering.*
(iii) $\mathcal{S}_H$ has a semi-conservative WNU*-polymorphism.*

Proof. Assume that H has no invertible pairs or chains. As it has no invertible pairs, its underlying graph H^R has none, so by Theorem 2.10, has a (parity-symmetric bipartite-)min ordering. Theorem 6.1, proved in the next section, shows that this ordering is special. So we get $(i) \Rightarrow (ii*)$.

Corollary 5.8, and Fact 5.4 proved in this section, give $(ii*) \Rightarrow (iii)$, and $(iii) \Rightarrow (i)$ respectively. □

The following actually gives implication $(iii) \Rightarrow (i)$ of the more general Conjecture 1.3. Is immediate as the underlying subgraph of a pr-graph may be assumed to be its red subgraph.

Fact 5.4. *If H is a* pr*-graph such that $\mathcal{S}_H$ has a* WNU*-polymorphism, then H has no invertible pairs or chains.*

Proof. Assume that $\mathcal{S}_H$ has a semi-conservative WNU-polymorphism. Then the underlying graph H^R of H has a WNU-polymorphism, and so by Theorem 2.10 it has a bipartite-min ordering, and so no invertible pairs. In [1] it is shown that chains in H omit semi-conservative WNU-polymorphisms on $\mathcal{S}_H$, so it has no chains either. □

Corollary 5.8 is an immediate corollary of Theorem 5.7. The proof of these are not long, but first we need a little understanding of the structure of special bipartite-min ordered pr-graphs.

A *red component* R of a pr-graph H is the subgraph induced by a set of vertices that are connected with red edges. We note that a red component may induce purple edges, but is connected via red edges. A neighbour of a vertex is called *purple* or *red* if it is adjacent via a purple or red edge.

Lemma 5.5. *Let H be a bipartite* pr*-graph having a special bipartite-min ordering, and let R be a red component of H with maximum vertices u_a and u_b. Let $r_a \sim r_b$ be a red edge in R, and let $x_a \sim x_b$ be an edge with $r_a \le x_a \le u_a$ and $r_b \le x_b$. Then $x_a \sim x_b$ is red, and x_a and x_b are in R.*

Proof. First we prove the lemma under the assumption that $x_b \le u_b$. As R is a red component there is a red path from r_a to u_a, and somewhere it must cross $x_a x_b$ so we may assume we have a red path aba' with $a \le x_a < a'$ and $b \le x_b$. By the

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x-underbar property we have $x_a \sim b$. As b is special $x_a \sim b$ must be red, and as x_a is special $x_a \sim x_b$ must be red, and in R.

Now throw away the assumption that $x_b \leq u_b$. If x_a is in R then $x_a x_b$ is red by x-underbar, and if x_a is not in R then $x_a x_b$ crosses some edge $r_a y_b$ of R with $y_b \leq x_b$ which says that $x_a \sim y_b$ by x-underbar. So x_a is in R by the previous case of the lemma. □

This shows that if there is a purple edge $x_a \sim x_b$ above a red edge $r_a \sim r_b$, then it must be above the whole red component R containing r_a and r_b. Or equivalently it shows the following which we will use in the next section.

Corollary 5.6. *If $r_a \sim r_b$ is a red edge and $x_a \sim x_b$ is a purple edge with $r_a < x_a$ and x_a being in the same red component as r_a, then $x_b < r_b$.*

We are now ready to prove the main result of the section.

Theorem 5.7. *Let $\mathcal{B}_H$ be a bipartite* pr*-graph with a special bipartite-min ordering $\leq$. The switch graph $\mathcal{S} := \mathcal{S}_{\mathcal{B}_H}$ has a semi-conservative* WNU*-polymorphism Φ. Moreover, if $\leq$ is parity-symmetric, then so is Φ.*

Proof. For a vertex v of $\mathcal{B}_H$ let $\mathrm{rc}(v)$ be the *red component* containing v. By Claim 4.2 it is enough to define $\Phi : \mathcal{S}^3 \to \mathcal{S}$ on constant parity tuples. For such tuples we set

$$\Phi((a_1, x_1), (a_2, x_2), (a_3, x_3)) = (A, X)$$

where $A = \min(a_1, a_2, a_3)$ for the bipartite-min polymorphism min defined by $\leq$, and X is defined as follows. Calling (a_i, x_i) or simply a_i *relevant* if $\mathrm{rc}(a_i) = \mathrm{rc}(A)$, let

- $X = \mathrm{maj}(x_1, x_2, x_3)$ if all a_i are relevant; otherwise
- $X = x_i$ where i is the minimum i such that a_i is relevant.

The function Φ is clearly idempotent, and semi-conservative. To see that it is WNU, it is enough to show that A and X are both WNU. A clearly is, to see that X is too, we assume that there is a repeated value in $\{a_1, a_2, a_3\}$ and observe that the value of X is chosen from among $\{x_1, x_2, x_3\}$ depending on which of the a_i is repeated, and is unchanged under permuting the indices $i = 1, 2, 3$. Indeed, if the repeated a_i is not relevant then X is the other x_i. If the repeated a_i is relevant, while the other is not, X is the repeated x_i. If all are relevant then X is majority, and so is the repeated x_i. In all cases, X is WNU. Observe also that Φ is switch-symmetric, as complementing the x_i does not effect A, and so does not effect which x_i is returned by X.

What is left to be shown, for the first statement of the theorem, is that Φ is a homomorphism. As it is switch-symmetric, by Lemma 3.1 it is enough to show that it preserves (not necessarily pure) red edges. Assume that $(a_i, x_i) \sim (b_i, y_i)$ is a red (or purple) edge for $i = 1, 2, 3$. We show that

$$(A, X) := \Phi((a_1, x_1), (a_2, x_2), (a_3, x_3)) \sim \Phi((b_1, y_1), (b_2, y_2), (b_3, y_3)) =: (B, Y),$$

is a red edge (or purple) edge.

Indeed, as min is a polymorphism of the underlying graph $\mathcal{B}_H^+$ of $\mathcal{B}_H$, we have

$$A = \min(a_1, a_2, a_3) \sim \min(b_1, b_2, b_3) = B$$

is an edge of $\mathcal{B}_H^+$, so is a red (or purple) edge of $\mathcal{B}_H$. If it is purple then whatever X and Y are, we have $(A, X) \sim (B, Y)$ is a red (or purple) edge, and we are done. So we must show that if $A \sim B$ is pure red then $X = Y$.

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If a_i is relevant, then $a_i \geq A$ and a_i is in $\mathrm{rc}(A)$. Any purple neighbour of a_i must be below B by the fact that a_i is special and by Lemma 5.5, so $a_i \sim b_i$ must be a red edge, making b_i relevant to B. The argument in the other direction is the same.

For the 'Moreover' statement of the theorem, we have that A is parity-symmetric by definition, as $\leq$ is parity-symmetric, so it is enough to observe that X is unchanged by replacing the a_i with $p_s(a_i)$. As A is parity-symmetric, doing so does not change which a_i are relevant, so this is immediate from the definition of X. □

The reason for the 'Moreover' statement in Theorem 5.7 is that it yields us the following corollary via Lemma 4.1.

Corollary 5.8. *If a* pr*-graph H has a parity-symmetric bipartite-min ordering then $\mathcal{S}_H$ has a semi-conservative* WNU*-polymorphism.*

6. Dichotomy for reflexive pr-graphs

To complete the proof of Theorem 3.2 we must show that following.

Theorem 6.1. *A reflexive* pr*-graph H with a min ordering has a special min ordering if and only if it has no chains.*

To prove this, we prove the following stronger statement, which allows us to use induction.

Theorem 6.2. *Let H be a* pr*-graph with a min ordering and no chains, and let H^0 be the purple graph with the same underlying graph. There is a sequence of $H^0, H^1, \ldots, H^b = H$ of* pr*-graphs with special min orderings such that for all $i \geq 1$, we get H^i from H^{i-1} by changing one purple edge to red.*

This clearly implies Theorem 6.1.

It helps to have a bit of an idea about what the typical special min ordered graph looks like. Referring to Figure 3 we point out some features, without proof. What we need, we will define and prove rigorously later. The graph in the figure is reflexive, though loops are not shown. A min ordered graph has a 'stem' subgraph consisting of a path with some extra x-underbar respecting edges, shown in the middle of the figure, with 'branches' shown above or below the stem, and not really drawn in their proper order. Each branch is itself an interval graph, and is dominated by its highest (right-most) neighbour on the stem: its 'branch-point'. Under a special min ordeing, if an edge is red, edges above it are red, but red edges in branches only force edges above them in their branch to be red. So there is generally one red component with edges in the stem, and it is an edge-upset. All other red components are in branches, and are connected to the stem via purple edges.

We start with some technical observations about min orderings before we start the proof.

6.1. Re-orderings of special min orderings of reflexive pr-graphs.

For any ordering $\leq$, we use $<$ for the standard 'strict' variation on the ordering. Recall that b *covers* a means that $a < b$ but that there is nothing strictly between a and b. An *up-neighbour* of a vertex is a neighbour that is above it, a *down-neighbour* is a neighbour that is below it.

We now give some technical lemmas which give conditions under which we can *re-order* H: make changes to the given min ordering and still have a min ordering.

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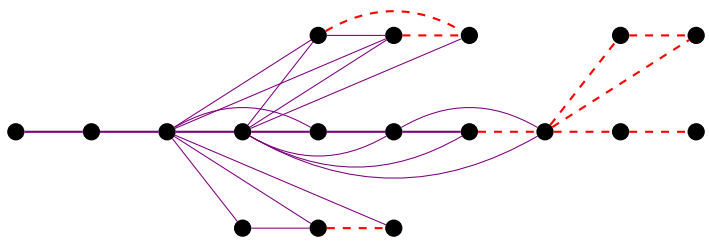

FIGURE 3. Typical special min ordered graph

These results can be proved using only the underbar identity (2), but another well known characterisation of min orderings yields faster and more intuitive proofs, so we first recall this.

A reflexive graph H is an *interval graph* if for each vertex b there is a real interval $I_b = [L_b, R_b]$ with $L_b < R_b$ (under the usual ordering of the reals) such that $a \sim b$ if $I_a \cap I_b$ is non-empty. It is well known and easy to see that (because H is finite) the endpoints L_b and R_b for all $v \in V(H)$ can be assumed to be distinct; and because of this that a reflexive graph H has a min ordering if and only if it is an interval graph. Indeed, if H is an interval graph the ordering of the left endpoints L_b induces a min ordering of H called the *left endpoint ordering*. On the other hand, given a min ordering $\leq$ of H, let the left endpoints be such $L_a < L_b$ if $a < b$; it is easy to find (usually several) interval representations of H using these left endpoints.

Reversing such an interval representation of $\leq$ by replacing each L_b with $-L_b$ and R_b with $-R_b$ gives another interval representation of H, and its left endpoint ordering is another min ordering of H. We call this an *essential reverse* of the original min ordering $\leq$ of H. Though an essential reverse ordering of a min ordering is not generally unique, so not just the reverse of the original ordering, the following is clear as non adjacent vertices have disjoint intervals.

Fact 6.3. *If a and b are non-adjacent vertices in H with $a \leq b$ under a min ordering $\leq$, then $b \leq a$ under any essential reverse of $\leq$.*

We will use the essential reverse ordering, but usually not on the whole graph at once. And we will use it in conjunction with other re-orderings, which we define now.

The following shows that we can move a above an adjacent cover b unless it has some neighbour below b that is not a neighbour of b.

Lemma 6.4. *Let H be a reflexive graph with $a \sim b$ and $\leq$ be a min ordering of H in which a covers b. If every neighbour c of b that is below a is also a neighbour of a, then there is a re-ordering of H that is the same as $\leq$ except that a covers b.*

Proof. As $a \sim b$ and $b \leq a$, in any L-representation of $\leq$ we have $L_b < L_a < R_b$. As a covers b there is no other vertex z with L_z between L_b and L_a. So moving L_b up until it passes L_a does not change any intersections unless there is some vertex c with $L_b < R_c < L_a$. Such c is adjacent to b and not a, and is below a, as needed □

We need a similar result for when a covers a non-adjacent neighbour b. In this case L_b and L_a do not intersect, so we will have to completely move L_b above L_a. We can do this if they have all the same neighours. So we also need that a has no

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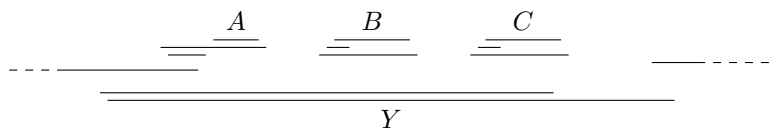

FIGURE 4. Branches A, B, and C. B and C are neat branches with container Y.

up-neighbours that b does not. However, we will need this in much more generality, telling us when we can move some set B containing b above some set A containing a. We start with several definitions.

Given an min ordering, a subset B of vertices of H is a *branch* if: it induces a connected subgraph, no vertex in B has an up-neighbour not in B, and all vertices of B have a common down-neighbour. The highest common neighbour x_B strictly below B is the *branch-point* of B. If there are any vertices above B, then the lowest vertex above B must also be adjacent to x_B. A branch B is *neat* if it satisfies the following for all $v \notin B$:

$$N(v) \cap B \in \{\varnothing, B\}. \tag{5}$$

More generally, without respect to a min ordering, a set B of vertices of H is *neat* if for all $v \notin B$ (5) is satisfied. For a neat set B, the set $Y = \{v \notin B \mid N(v) \cap B = B\}$ is its *container*. We say Y *contains* B.

See Figure 4. The three-vertex subgraphs labelled A, B and C are all branches, but only B and C are neat, both having container Y.

Given a branch B of H the vertices Y in $H \setminus B$ with edges to B are all below B, so induce a clique by the underbar condition. If they all dominate B then B is a neat branch.

A neat set need not be a branch under a given min ordering, as it may contain vertices that are below vertices in its container. However as long as its container is a clique, there is a simple re-ordering that makes it a neat branch. For this, we need another definition.

A clique Y in a min ordered graph H is *separated* if there is an interval s of the real line such that the vertices $v \in V(H)$ for which $s \subset I_v$ are exactly those in Y. Such an interval s is called a *separator* for Y. It is a well known fact about interval graphs that maximal cliques are separated under an interval ordering.

Fact 6.5. *Let Y be the container in H of a neat set B, and let $\leq$ be any min ordering of H. If Y is a clique, then it is separated in the restriction of $\leq$ to $H \setminus B$.*

Proof. As Y is a clique and is the container of B there is some maximal clique of H containing only vertices in $Y \cup B$. This has some separator s which in $H \setminus B$ is then a separator of Y. □

For a neat set B of H whose container Y is a clique, the fact that H has a min ordering tells us that Y has a separator s, and so clearly one can remove B and re-insert it in s getting a re-ordering in which B is a neat branch. As we always assume that H has a min ordering, we call neat set B whose container is a clique a *neat branch*. This same reordering can be done more generally.

Lemma 6.6. *Let $B_1, \ldots, B_r$ be neat branches of H with containers $Y_1, \ldots, Y_r$. From any min ordering $\leq$ of H, and any min ordering $\leq_i$ of B_i, there is a re-ordering $\leq'$ of H such that:*

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(i) $\leq'$ *agrees with* $\leq$ *on* $H \setminus \bigcup B_i$,
(ii) $\leq'$ *agrees with* $\leq_i$ *on* B_i,
(iii) the vertices of B_i *occur consecutively and above* Y_i,
(iv) if $Y_i = Y_j$ *then* $B_i < B_j$. *(This allows an arbitrary ordering of neat branches with the same container.)*

Proof. From the min ordering $\leq$ of H, remove all of the B_i. Ordering them by the minimum element of their containers, take the lowest one B_i and insert it, with ordering $\leq_i$ in the highest separator of its container, which exists by Fact 6.5. □

6.2. **Common chains and notation to construct them.** Recall from Definition 5.2 that a chain is defined as a directed walk

$$(a_1, b_1) \to \cdots \to (a_d, b_d)$$

in $\mathcal{A}$ from a good end (a_1, b_1) to bad end (a_d, b_d). This translates to a pair of walks in H between some u and v. We denote them as

$$u \left| \begin{array}{c} a_1 a_2 \dots a_d \\ b_1 b_2 \dots b_d \end{array} \right| v$$

where the important properties are

- $ua_1a_2 \dots a_d y$ and $xb_1b_2 \dots b_d v$ are walks,
- ua_1 and $b_d y$ are red while xb_1 and $a_d v$ are purple, and
- for $i = 1 \dots, d-1$ we have $b_i \not\sim a_{i+1}$.

Often walks in the chain will have repeated vertices, and we will construct them by concatenating predefined paths. Extending this notation, we might write something such as the following:

$$u \left| \begin{array}{c} Pa \\ b^* \end{array} \right| \begin{array}{c} a^*c \\ Q \end{array} \left| v \right. \qquad \text{or} \qquad u \left| \begin{array}{c} Pa \\ b^* \end{array} \right| \begin{array}{c} a^*c \\ Q \end{array} \left| \right.$$

Here P and Q are defined walks, writing them in sequence denotes concatenation, so, for example, Pa is the walk P extended by the vertex a. The notation x^* denotes the walk that visits the vertex x repeatedly however many times is required so that the two walks between the separators '|' have the same length. If the final vertex v is missing from our notation, as in the second example, it is because the two walks, a^*c and Q in the above example, both end at the same vertice, and it is playing the role of v.

There are a couple of chains that we use so often that we give them names. A min ordering of H is a bipartite min ordering of $K_2 \times H$, so a chain in $K_2 \times H$ obstructs min orderings in H.

A *prp-chain* is any walk $v_1v_2 \dots v_\ell$ in H with $\ell \geq 4$ such that the first and last edges are purple, the others are red, and for all i we have $v_i \not\sim v_{i+2j}$ for any $j \geq 1$. This yields an induced path in $K_2 \times H$ in which just the first and last edge are purple. This was observed in [1] to be a chain in $K_2 \times H$ and so to obstruct bipartite-min orderings in $K_2 \times H$. So it obstructs min orderings in H.

Similarly an *rpr-chain* is a walk $v_1v_2 \dots v_\ell$ in H with $\ell \geq 5$ such that the first two and last two edges are red, the others are purple, and for all i we have $v_i \not\sim v_{i+2j}$ for any $j \geq 1$. This also induces in $K_2 \times H$ a chain given in [1].

An *alternating* C_4 is a four cycle $v_1v_2v_3v_4$ with red edges v_1v_2 and v_3v_4 and blue edges v_2v_3 and v_4v_1 not necessarily induced, and having loops of any colour

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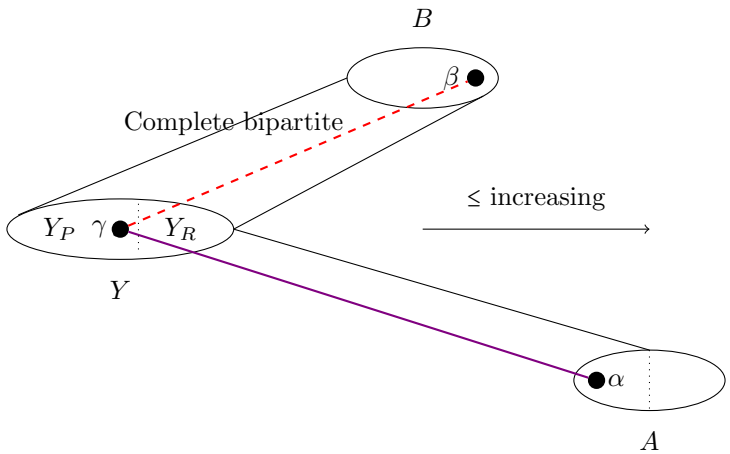

FIGURE 5. The structure of H described in Lemma 6.7

contains the following chain:

$$v_1 \left| \begin{matrix} v_2 \\ v_4 \end{matrix} \right| v_3$$

6.3. **Proof of the main theorem.** Our proof of Theorem 6.2 is an induction on the number of red edges of H. The theorem is trivial for purple graphs having no red edges. We may assume that we get H from a previous graph H^p with a special min ordering $\leq$, by making a red edge $\gamma\beta$ purple.

As $\leq$ is still a min ordering of H, Theorem 6.2 is proved if it is special, so we may assume that making $\gamma\beta$ purple has introduced at least one *specialty violation* $(\gamma; \alpha, \beta)$: a vertex γ with a purple neighbour α above a red neighbour β. Our task it to re-order the vertices, removing this specialty violation, without introducing any new ones. If we can to this, we are done, again by induction.

We have $\beta \leq \alpha$ by definition, but we may have $\gamma < \beta$, $\gamma = \beta$ or $\gamma > \beta$. Lemma 6.7 deals with the latter two cases, and gathers together several other reductions, allowing us to assume a fair bit about the structure around γ, β and α. The description of this structure in the lemma is a bit technical, but many of its features are shown n Figure 5. Note that the figure might not show all vertices of the graph, but it does 'show' all adjacent to β.

After we prove this, we then have two main cases to consider: the case that Y dominates A, and the case that it does not.

Lemma 6.7. *We may assume that β is in a neat branch B with γ in its container Y, and that α (covering β) is in a different component A of the graph we get from H by removing Y and all vertices below Y. Moreover, we may assume that $Y \setminus \{\gamma\}$ splits into sets $Y_P < Y_R$ such that all edges from $B \cup A$ to Y_P are purple and all to Y_R are red.*

Proof. A speciality violation $(\gamma; \alpha, \beta)$ is of *type 1* if $\gamma \leq \beta$ and of *type* 2 if $\gamma > \beta$. Its *width* is the distance, with respect to the ordering, between α and β. H may have many speciality violation, but we commit to choosing one according to the following precedence:

- choose a type 1 speciality violation of minimum width, if one exists,

- otherwise choose a type 2 speciality violation of minimum width.

We now make a progression of claims that yield the lemma.

Claim 6.8. *We may assume there is no neighbour of γ between α and β.*

Proof. This is by the assumption that $(\gamma;\alpha,\beta)$ is a speciality violation of minimum width. Indeed, if there is another neighbour of γ between α and β, this would make a speciality violation with γ and either α or β, of lesser width. If $(\gamma;\alpha,\beta)$ is of type 1, then so is the new violation. This then contradicts our choice of $(\gamma;\alpha,\beta)$. ◇

Claim 6.9. *Any purple neighbour of β is a purple neighbour of α. In particular any neighbour of β below γ is a purple neighbour and adjacent to α.*

Proof. If β has a purple neighbour c that is not adjacent to α then $c\beta\gamma\alpha$ is a *prp*-chain in H. If it is adjacent but via a red edge, then we have an alternating C_4. If β has any neighbour below γ then by the speciality of β in the ordering on H^p, it is a purple neighbour. ◇

Claim 6.10. *We may assume that $(\gamma;\alpha,\beta)$ is of type 1.*

Proof. Assume that $(\gamma;\alpha,\beta)$ is of type 2. We will show that we may resolve the speciality violation, or find another speciality violation of type 1, contradicting our choice of $(\gamma;\alpha,\beta)$.

As we assume $\gamma > \beta$, by speciality in H^p the loop on β is purple. If the loop on γ is purple, then $\beta\beta\gamma\gamma$ is an alternating C_4, so it is red. As the edge from γ to α is purple, α is below γ by the speciality of γ in H^p, and has a purple loop by speciality and a purple edge to β by the underbar identity.

If β has a neighbour c below α that is not adjacent to α then it must be a red neighbour, or $c\beta\gamma\alpha$ would be a *prp*-chain. If there is such a red neighbour, then it is above β by speciality and $(\beta;\alpha,c)$ is a speciality violation of type 1, which contradicts our choice of $(\gamma;\alpha,\beta)$.

Thus we can assume that all neighbours of β are also neighbours of α. We claim we can now use Lemma 6.4 to get a re-ordering in which β covers α. This is immediate from the lemma if there are no vertices between α and β. If there are, let $c < \alpha$ be the cover of β. As α is adjacent to β and any of its neighbours below α, we have by the underbar identity that c is adjacent to β and any of its neighbours below c. So the lemma allows us to move β up past c. Moving, in the same way, past all intervening vertices we move β above α and end up with our re-ordering.

The speciality violation $(\gamma;\alpha,\beta)$ is resolved in the re-ordering. Assume some new speciality violation is introduced; as only β has moved past vertices $c < \alpha$, it is of the form $(x;\beta,c)$ for some purple neighbour x of β with a red edge to some c originally between β and α. If $x \sim \alpha$ then this is a red edge by speciality, and so α,γ,β,x is an alternating C_4, which is a chain. Thus $x \not\sim \alpha$, which means that $c \neq \beta$ and that $c \neq \gamma$ by Claim 6.8. But then

$$\beta \left| \begin{array}{c} \gamma\alpha \\ xx \end{array} \right| c$$

is a chain in H.

◇

Claim 6.11. *α covers β*

Proof. As $\gamma \leq \beta$ this is immediate from the underbar identity and Claim 6.8. ◇

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Claim 6.12. *$\gamma \neq \beta$, and so $\gamma < \beta < \alpha$*

Proof. Assume now that $\gamma = \beta$, so γ has a red loop and α is a purple neighbour above it. If γ has any other purple neighbours below α, they are thus below γ, and so by Claim 6.9 they form a purple clique with α. All red neighbours of γ (other than γ itself) are above these purple neighbours by the speciality of γ in H^p. By Lemma 6.4 we can move $\beta = \gamma$ above α.

If the re-ordering introduces a new specialty violation, it is of the form $(y; \gamma, c)$ for some purple neighbour c that γ moved above, and a vertex y with a red edge to c and a purple edge to γ. But then $\gamma\gamma y\alpha$ is an alternating C_4. We therefore assume that $\gamma \neq \beta$.

◇

Claim 6.13. *We may assume that $\alpha \not\sim \beta$ in H.*

Proof. If we cannot move β up past α then by Lemma 6.4 there is neighbour c of β that is below α (and above γ) and not adjacent to α. As α has a purple edge to γ, so does β. But then

$$\gamma \left| \begin{array}{c} \beta c \\ \alpha\beta \end{array} \right| \gamma$$

is a chain in H. ◇

As α covers β but they are not adjacent, there are no edges from β to above α. Let Y be the set vertices strictly below β with neighbours strictly above β; Y is non-empty as it contains γ. Where $D(Y)$ is the set of vertices below the maximum element of Y, let A and B be the components of $H \setminus D(Y)$ that contain α and β respectively. By construction it is evident that B is dominated by all vertices of Y, so is a branch whose branch point is the maximum vertex of Y. To finish off the first statement of the lemma, we show that B is neat.

Claim 6.14. *We may assume that any neighbour $c \notin B$ of any vertex in B is adjacent to α, and so (being below B) dominates B. Thus B is neat.*

Proof. Let $c \notin B$ be a neighbour of B that is not adjacent to α, so is not in Y. If $\beta \sim c$ then this is a purple edge, and $c\beta\gamma\alpha$ is a *prp*-chain. So we may assume that $\beta \not\sim c$.

Now observe that c dominates Y. Indeed, any element y of Y that is above c is adjacent to it by underbar using their edges to β, while any element of y below c is adjacent to it by underbar and the edge αy, so c dominates Y.

Set $C = \{c\}$ and expand C as follows. While there is z in C with a neighbour d below B but not in Y, add d to C. We claim that such d also dominates Y.

Indeed, if it did not, we would have the following chain in H, where P is a shortest path from β to z in $B \cup C$:

$$\gamma \left| \begin{array}{c} P \\ \alpha^* \end{array} \right| \begin{array}{c} dd\gamma \\ z\beta\beta \end{array} \left| \gamma. \right.$$

Thus $C \cup B$ is connected, and for all vertices in C the neighbours not in $B \cup C$ are also exactly those in Y. So $C \cup B$ is a neat set contained in a clique. Applying Lemma 6.6 we put all of $B \cup C$ just above the maximum element of Y as a neat branch.

We now verify that no new speciality violations were introduced. The only vertices that moved are those in C, and they only moved relative to neighbours in

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Y all of which were purple neighbours. All vertices in Y were special in H^p and had purple neighbours to β which is above the new position of all vertices in C, so the vertices in Y (except possibly γ, whose violations have been reduced) remain special. Vertices in C have had purple neighbours move relatively down, so they also remain special. ◇

The following will give us the 'Moreover' statement of the lemma. We also use it again in a later case.

Claim 6.15. *There is no vertex $y \in Y$ with a purple edge to B and a red edge to A.*

Proof. If y has purple neighbour β_0 in B an red neighbour α_0 in A then where P_B is a path from β to β_0 in B and P_A is a path from α to α_0 in A, the following is a chain:

$$\gamma \left| \begin{matrix} P_A \\ P_B \end{matrix} \right| y$$

□

It follows by speciality that any $y \in Y \setminus \{\gamma\}$ has only red or purple edges to $A \cup B$. Let Y_R consist of the vertices of $Y \setminus \{\gamma\}$ with red edges to $A \cup B$ and Y_P those with purple edges. Again by speciality all of Y_R is above $Y_P \cup \{\gamma\}$ and if γ has any red edges to A then it is above Y_P. □

Now that Lemma 6.7 is proved, we have two cases to consider. If A is also a branch of Y, then we find a local re-ordering allowing us to move B above A. If A is not a branch then a local reordering will not do, we must then show that we can take an essential reverse ordering of H to move B above A. We have to do this without disrupting speciality everywhere, so have to show that all red edges are in neat branches, and so we can keep the original orderings within the branches.

Case 1: *All vertices in A also dominate Y.* Let H' be the subgraph induced by $A \cup B \cup Y_R^+$ where $Y_R^+ \supset Y_R$ is all vertices strictly below B but not strictly below Y_R. Vertices in Y_R^+ dominate $Y_P^\gamma := Y_P \cup \{\gamma\}$ by the underbar condition, because B is above Y_R^+ and dominates Y_P^γ. Furthermore, $A \cup B$ dominate Y by assumption, so H' is a neat branch contained in Y_P. It follows by Lemma 6.6 that any re-ordering of H' leaving the rest of H unchanged, is a new min ordering of H.

Within H', we claim that we can switch the relative order of A and B keeping the ordering within each of them the same, and still have a min ordering. Indeed if Y_R is empty, this is trivial, as then they are separate components of H'; and if Y_R is not empty then they are separate neat branches contained in Y_R, so the claim follows by Lemma 6.6.

We claim that the new min ordering this defines on H is special. Indeed it is enough to verify that vertices in Y are special, as any other vertices were already special and none of their neighbours have been reordered relative to each other. For all vertices in Y except γ all edges to $A \cup B$ are the same colour, so as the vertices were special before the re-ordering, they are special after the re-ordering.

If the only red neighbour of γ is β, then because β is now the maximum vertex, γ is special. So we may assume that γ has some red edge to A. In this case $B = \{\beta\}$, as otherwise, any other vertex of B would have a purple edge to γ and contradicting Claim 6.15. Thus moving B above A moves only the one red neighbour β to the

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top of its neighbourhood, and γ again becomes special. $\diamond$

This completes the proof for Case 1, so we can assume that there is some vertex in Y that is not adjacent to some vertex in A. By the underbar property, the vertex in A may be assumed to be the top vertex of A. We are left with the following case.

Case 2: *Some vertex in Y is not adjacent to the maximum vertex a_M of A.* We start with several quick observations.

Lemma 6.16. *The following all hold.*

- *(i) Neighbours of γ in A dominate Y_P.*
- *(ii) Vertices in Y_P^γ are adjacent to all vertices in A that have red loops.*
- *(iii) Red neighbours of γ in A dominate Y.*
- *(iv) The vertex a_M has a purple loop.*
- *(v) Any red edge or loop in A is in a neat branch of A.*
- *(vi) All edges from Y to A are purple, so $Y = Y_P^\gamma$.*
- *(vii) Any vertex below B that is adjacent to but not in Y dominates Y.*
- *(viii) Any vertex below B that is adjacent to but not in Y is in a neat branch of H contained in Y.*

Proof. Later items use earlier items; we prove them in order.
(i) Towards contradiction, assume $\gamma \sim a \not\sim y$ for some $a \in A$ and $y \in Y_P$. The edge $\gamma\beta$ is always red and γa is purple by speciality as $\alpha \le a$. The edge $y\beta$ is purple by definition of Y_p. So the following is a chain:

$$\gamma \left| \begin{matrix} \beta y \\ a\gamma \end{matrix} \right| \beta$$

(ii) Assume some $y \in Y_P^\gamma$ has a non-neighbour $a \in A$ with a red loop. By item (i) $\gamma \not\sim a$. Let P be a shortest decreasing path in $A \cup \{\gamma\}$ from a to γ. (By speciality, as soon as one edge of a decreasing path is purple, all edges that follow it are also purple). Let s be the last vertex in P with a red edge in P, or, if P is purple, let $s = a$. Augment P to a walk W of the form a^*P or $P\gamma^*$ having s as its middle vertex. Let W_1 be the first half of W and W_2 the second. The following is then a chain (notice that $(W_2)^{-1}$ and W_1 both end at s):

$$\gamma \left| \begin{matrix} \beta \\ \alpha \end{matrix} \right| \begin{matrix} \beta^*\gamma \\ P^{-1} \end{matrix} \left| \begin{matrix} (W_2)^{-1} \\ W_1 \end{matrix} \right.$$

(iii) For $a \in A$ let γa be red and assume $a \not\sim y$ for some $y \in Y$. By item (i) y must be in Y_R, so $y\alpha$ is red. Let P_A be a path in A from α to a. The following is a chain:

$$\gamma \left| \begin{matrix} \beta^* \\ P_A \end{matrix} \right| \begin{matrix} y \\ a \end{matrix} \left| \gamma \right.$$

(iv) Assume that a_M has a red loop. By item (ii) we have $\gamma \sim a_M$, and so by item (iii) we have that a_M dominates Y. This contradicts the case assumption.

(v) As A is connected and a_M has a purple loop by item (iv), any red edge must be in a branch C of A that has purple edges to its branch point a_C. Let a_u be the lowest neighbour of a_C above C in A. If C is not neat, there is some vertex v below a_C in A that is adjacent to some non-maximum c_d in C but to no c_u above c_d in C.

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Let Q be any path in A from α to a_M, and let R be a shortest decreasing path in A from a_M to a_C. (We know that only a_C in R can be adjacent to v.) Let P be a shortest decreasing path in $C \cup \{a_C\}$ from c_u to a_C. Make W with middle vertex s the transition from red to blue by prepending P with c_u^* or appending it with a_C^* and let W_1 and W_2 be the first and last halves of W.

The following is then a chain:

$$\gamma \left| \begin{matrix} \beta^*\gamma \\ Qa_M \end{matrix} \right| \begin{matrix} v^* \\ R \end{matrix} \left| \begin{matrix} v \\ c_u \end{matrix} \right| \begin{matrix} (W_2)^{-1} \\ W_1 \end{matrix}$$

(vi) If there is some $y \in Y_R$ then it has a red edge up to α, and so α has a red loop. Being the minimum vertex of the connected graph A, A is then all red by speciality, and in particular a_M has a red loop, contradicting item (iv). So Y_R is empty. We just have to show now that γ has no red-neighbour a in A. If it did, then a has a red loop, so by item (v) a has an attachment a_C in A to which it has a purple edge. But a has a red edge to γ, which is below a_C, so this would contradict the speciality of a.

(vii) Let $y_\ell \sim c \not\sim y_u$ where $c < B$ and $y_\ell, y_u \in Y$. By the underbar identity we have $y_\ell < y_u$, (if c is not below both of them this uses the edge from $\beta > c$ to y_ℓ). The following is a chain where P is a path in A from α to a_M, Q is a shortest path from γ to c, R is a path descending from from a_M to y, and S is a path from c upto γ:

$$\gamma \left| \begin{matrix} \beta^* \\ P \end{matrix} \right| \begin{matrix} Q \\ a_M^* \end{matrix} \left| \begin{matrix} c^* \\ R \end{matrix} \right| \begin{matrix} S \\ \beta^*\gamma \end{matrix}$$

(viii) Let $c \notin Y$ be below B and have a neighbour $y \in Y$. By item (vii) it is enough to show that if $c \sim c'$ for some $c' \notin Y$ also below B, then c' also has a neighbour in Y. If it doesn't then where a is some least element in A not adjacent to γ, the following is a chain:

$$\gamma \left| \begin{matrix} \beta\beta \\ \alpha a \end{matrix} \right| \begin{matrix} \gamma c \\ \alpha\gamma \end{matrix} \left| \begin{matrix} c'c \\ \beta\beta \end{matrix} \right| \gamma$$

□

As Y has no edges to $a_M \in A$, it has no edges to above A. Any neighbours of Y between a_M and α have edges to α by underbar, so are in A. Any neighbours below β but not below B are similarly in B. So the only neighbours Y has not in $A \cup B$ are below B, and by Lemma 6.16(vii) are in neat branches contained in Y. Let $\mathcal{B}_Y$ be the set of these neat branches contained in Y. Thus H consists of $A \cup B \cup Y \cup \bigcup \mathcal{B}_Y$.

B is a neat branch, by 6.16(v) all red edges in A are in neat branches, the collection of which we call $\mathcal{B}_A$, and as $Y = Y_P^+$ all edges from Y to branches in $\mathcal{B}_Y$ are purple. So the only red edges in H are $\beta\gamma$ or in B or in neat branches in $\mathcal{B}_A$ or $\mathcal{B}_Y$.

Apply Lemma 6.6 with the an essential reverse ordering on H and with the original ordering on the set $\mathcal{B} := \mathcal{B}_A \cup \mathcal{B}_Y \cup \{B\}$ of neat branches, the order of the neat branches is unimportant except that B should be the last one. Observe that the clique Y is at the top of the essential reverse ordering of $H \setminus \bigcup \mathcal{B}$. Moreover, by Lemma 6.16 parts (i) and (vii) the vertices of Y all have the same neighbours, so we may assume that γ is the top vertex in Y. As B is the last neat branch, it

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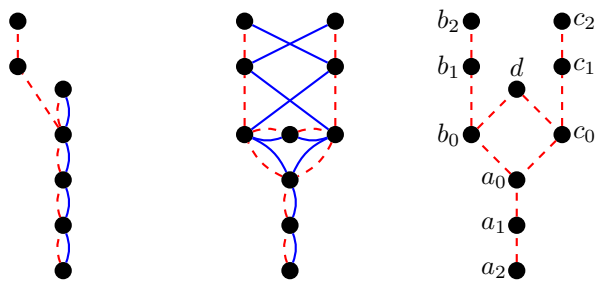

FIGURE 6. The tree T, the switch-core of $\mathcal{P}_T$, and its red subgraph R

is the highest ordered of all branches with container Y, any branches with different containers have containers properly containing Y, so are ordered below B by underbar. Thus β is the maximum element of the re-ordering.

This re-ordering resolves the speciality violation $(\gamma; \alpha, \beta)$, indeed γ becomes special as its only red neighbour β is now the maximum vertex. To see that it add no other speciality violations observe that any vertex with a red edge, except β and γ, is in a neat branch in $\mathcal{B}$, and all edges to its container are purple, so these vertices are special as they were before the reordering. The only possibly non-special vertex is β, but the only of its neighbours that have moved relative to each other are those in $Y \setminus \{\gamma\}$, and all of these are purple neighbours. So any speciality violation at β was already there.

7. Possible and impossible polymorphism collapses

The following questions arise naturally when considering the complexity dichotomy for $\mathsf{LSwHom}(H)$, and when observing the various polymorphism collapses we have mentioned in the paper.

Question 7.1. *Is it true for a* br*-graph H that*

(i) $\mathcal{S}_H$ has a semi-conservative WNU *if and only if it has a parity-symmetric one?*

(ii) $\mathcal{S}_H$ has a semi-conservative WNU*-polymorphism if and only if it has a conservative* WNU*-polymorphism?*

(iii) $\mathcal{S}_H$ has a semi-conservative WNU*-polymorphism if and only if it has a semi-conservative* 3-NU*-polymorphism?*

Part (i) is certainly tempting. Using the results and ideas of Section 4 it is the key to getting a full reduction to the bipartite case: $\mathsf{LSwHom}(H) =_{\text{poly}} \mathsf{LSwHom}(\mathcal{B}_H)$. Part (ii) was tempting, reducing the whole paper to Section 3, and (iii) also, as it would reduce the paper to Sections 3 and 4. We now have countless counterexamples giving negative answers to these last two questions. We give one in the proposition below.

Proposition 7.2. *Where T is the* br*-graph on the left of Figure 6, $\mathcal{P}_T$ has a semi-conservative* WNU*-polymorphism but no conservative* WNU*-polymorphism or semi-conservative* 3-NU*-polymorphism.*

Proof. That $\mathcal{P}_T$ has a semi-conservative WNU-polymorphism is shown in [1], and also follows from Theorem 5.7. Indeed, ordering the vertices of T according to their

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position in the figure as one moves up the page; except putting the top vertex in the purple path below its neighbour, we have a special bipartite-min ordering of T.

It is not too hard to verify that the triple (a_2, b_1, c_1) in R is a permutable triple, showing that it and so $\mathcal{P}_T^R$ (of which it is an induced subgraph) can have no conservative WNU-polymorphism.

We show now that R can have no semi-conservative 3-NU-polymorphisms. Note that now symmetric lists are those for which if b_i is in a list then c_i is also, and vice-versa.

Assume, towards contradiction, that ϕ is a semi-conservative 3-NU-polymorphism of R. There is no valid image for $\phi(a_2, b_1, c_1)$. Indeed, if $\phi(a_2, b_1, c_1) = a_2$, then $\phi(a_1, b_0, c_0) = a_1$, and $\phi(a_2, d, d) = a_2$. This contradicts the fact that ϕ is a 3-NU. So without loss of generality we may assume that $\phi(a_2, b_1, c_1) = b_1$, Then $\phi(a_1, b_2, c_2) = b_2$, $\phi(a_0, b_1, c_1) = b_1$, $\phi(c_0, b_2, c_2) \in \{b_0, b_2\}$ and $\phi(c_1, b_1, c_1) = b_1$, again contradicting the fact that ϕ is a 3-NU. □

In [1] the authors observe that there are infinitely many trees having semi-conservative WNU-operations that contain T as an induced subgraph. The proof works for all of these.

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